Vina–RTMScore hybrid docking

Standalone inference · rigid receptor · CPU local optimization + GPU neural scoring

Receptor PDB + ligand SDF / MOL2 + binding region

Manual box center / size, or a bound reference ligand defines the box

CPU

Input preparation and atom mapping

Independent 3D start · PDBQT · chemical template / atom map · protein pocket

Vina-only warmup, then RTMScore evaluation

Local Vina relaxation; warmup Metropolis decisions use Vina energy E
RTMScore evaluates all valid warmup candidates

Fix normalization from the warmup candidate pool

Median centers and IQR scales for E and S; scale floor = 0.001
Calibration remains fixed throughout hybrid search

Python multichain / multistart search

Rigid-body and active-torsion proposals

Default budget: 32 chains × 256 steps × 4 starts

CPU

Vina local BFGS minimization

Independent CPU states · cached radial tables · optimized coordinates + E

Chemical identity and boundary checks

Reject changed stereochemistry, open macrocycles or out-of-box poses

GPU

Batched RTMScore inference

Atom / residue graphs → cached neural mixture coefficients
Optimized coordinates → distances → probability density → score S

RTMScore S

Hybrid objective: $J = z_E - \lambda z_S$

Fixed warmup normalization; default λ = 1; lower J is better

Metropolis acceptance using J

Accept: update parent pose; reject: retain parent pose
Default temperature T = 1

Next step: updated / retained parent state

Vina energy E

After all starts

Sort the archive of all valid candidates by J

Includes warmup candidates and accepted / rejected hybrid proposals
Invalid candidates are excluded

Top-1 / Top-N poses + score table

Vina energy E · RTMScore S · hybrid objective J

RTMScore controls global acceptance and final ranking; Vina local BFGS uses Vina energy only.

Solid arrows: computation / data flow. Dashed arrow: parent-state feedback. No RTMScore gradient enters Vina.